



SehatAI

Rural Health Risk & Triage Copilot

Project Report

PakAngels GenAI & Agentic AI Training — Cohort 11
In collaboration with PakAngels, iCodeGuru & ASPIRE Pakistan

Prepared by Team SehatAI

1. Jahangeer Ali (Team Lead)
2. Kheem Pakrnash Dharmani
3. Maria Talib ·
4. Huda Tahir
5. Hajra Arshad
6. Saira Lateef

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1. Executive Summary

SehatAI is an AI-assisted health screening and patient-education prototype built for rural and resource-constrained communities. The application combines two machine-learning risk-screening models (for diabetes and cardiovascular disease), an OCR-based medical report reader, and a bilingual conversational assistant powered by a free-tier Groq large language model, all wrapped in a Streamlit web interface designed to be usable by people with little formal education.

The system offers a "Simple Mode" that only asks questions an ordinary person can answer about themselves (weight, height, an approximate blood-pressure reading, family history) and an "Advanced Mode" for clinics and health workers who have exact laboratory values. Both modes feed a shared Random-Forest based risk-scoring pipeline, and every result is explicitly presented as a screening estimate rather than a diagnosis, with clear guidance to consult a qualified healthcare professional. This report documents the problem, design, technology stack, datasets, model evaluation results, and the working application built during the PakAngels GenAI & Agentic AI Training hackathon.

2. Problem Statement

Access to basic preventive health screening remains a major challenge in rural and under-served parts of Pakistan and similar regions. Several compounding issues motivated this project:

- Specialist doctors and diagnostic laboratories are concentrated in urban centres, leaving rural clinics and villages with little access to timely screening.
- Patients who do obtain a lab report are often unable to interpret the numbers, units and medical terminology it contains.
- Chronic conditions such as diabetes and cardiovascular disease are frequently detected late, after symptoms have already become severe.
- Most existing digital health tools are English-only and assume a level of digital and health literacy that excludes a large share of the population.

SehatAI was designed to reduce this gap by giving patients — and the community health workers who support them — a friendly, bilingual, first-line screening companion that works with the information an ordinary person already has, rather than requiring a laboratory visit before any guidance can be given.

3. Proposed Solution & Objectives

SehatAI is a Streamlit web application that turns patient-entered vitals, free-text symptoms, and optionally an uploaded medical report into a plain-language screening summary, while keeping a human doctor in the loop for every important decision. The core objectives were:

- Build and evaluate machine-learning models that estimate diabetes and cardiovascular risk from routine clinical indicators.

- Make the screening form usable by people with little formal education, without sacrificing accuracy for clinics that do have exact lab values.
- Automatically read PDF or photographed lab reports and turn the numbers inside them into the same risk estimates and a plain-language explanation.
- Offer a conversational, empathetic health assistant that asks short follow-up questions before giving general, non-diagnostic guidance.
- Support English, Urdu and Roman Urdu throughout the interface and the AI-generated guidance.

How it Works

A patient — or a health worker on their behalf — fills in the Patient Screening tab (in Simple or Advanced Mode), optionally describes symptoms in free text, and optionally uploads a lab report. SehatAI aligns whatever values are available with the features the trained models expect, silently filling in any value the patient could not reasonably know with a population-typical default, then produces a diabetes-risk and a cardiovascular-risk percentage. A Groq-hosted language model turns those numbers, the reported symptoms, and any extracted report text into an organised, easy-to-read explanation — summary, what it means, healthy next steps, when to see a doctor, and emergency warning signs — always finishing with a clear, non-diagnostic disclaimer.

4. System Architecture

The diagram below summarises how a single screening request flows through SehatAI, from the patient's form entry to an optional human review step.

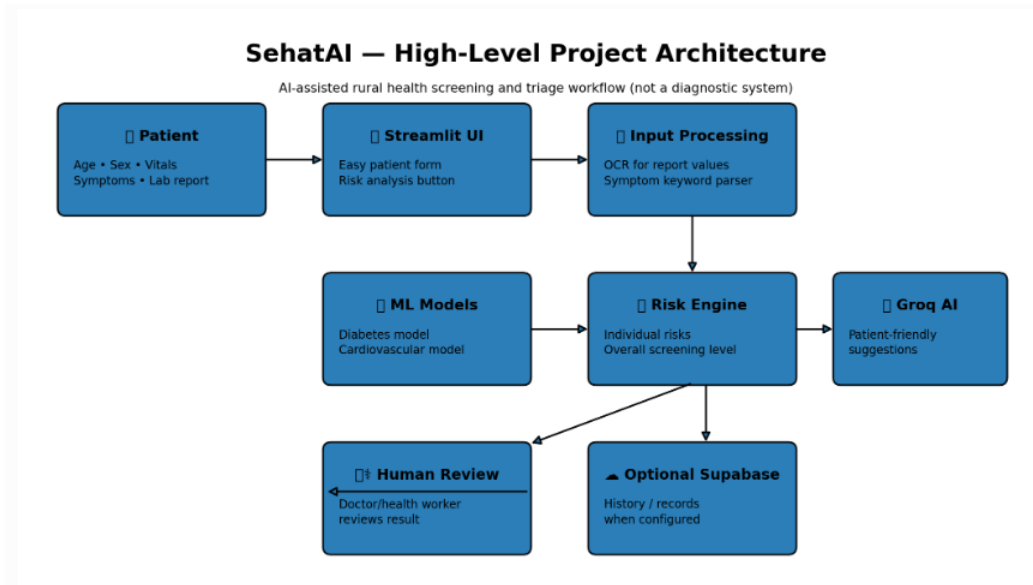


Figure 1 — SehatAI high-level architecture.

Patient input (age, sex, vitals, symptoms, and an optional lab report) is captured by the Streamlit UI and passed to an input-processing layer that performs OCR/text extraction on reports and keyword parsing on free-text symptoms. The resulting values are merged with the patient's form data and sent to the two trained ML models. The risk engine combines their outputs into individual and overall screening levels, which are handed to the Groq-hosted language model to produce patient-friendly suggestions. The final result is presented for human review by a doctor or health worker, with an optional history-storage layer (Supabase) for future longitudinal tracking.

5. Technology Stack

Layer	Tools & Frameworks	Purpose
Interface	Streamlit (Python), custom CSS	Bilingual, responsive web UI with Simple/Advanced screening modes and a sidebar chat assistant.
Machine Learning	scikit-learn (Random Forest), XGBoost, joblib	Training, comparison, and persistence of the diabetes and cardiovascular risk models.

Layer	Tools & Frameworks	Purpose
Explainability	SHAP (TreeExplainer)	Surfacing the top features behind an individual model prediction.
Conversational AI	Groq API — "openai/gpt-oss-20b" (free tier)	Symptom triage conversation and plain-language report/guidance generation.
Report Reading	pypdf, pytesseract, Pillow, regex	Extracting text from PDF/image lab reports and pulling out recognisable clinical values.
Data Handling	pandas, numpy	Loading, cleaning and preparing the Pima Diabetes and UCI Heart Disease datasets.
Visualization	matplotlib, seaborn	Model-comparison charts used during evaluation.
Deployment	GitHub, Streamlit Community Cloud	Version control and hosting of the live application.

6. Dataset Description

Two well-known, publicly available clinical benchmark datasets were used to train SehatAI's screening models.

Diabetes:
(768, 9)

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Heart Disease:
(303, 14)

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3	3.0	0.0	6.0	0
1	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5	2.0	3.0	3.0	1
2	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6	2.0	2.0	7.0	1
3	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5	3.0	0.0	3.0	0
4	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4	1.0	0.0	3.0	0

Figure 2 — First rows of the Diabetes and Heart Disease training datasets.

6.1 Pima Indians Diabetes Dataset

768 records and 9 columns: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, and the binary Outcome label. Biologically impossible zero values in Glucose, BloodPressure, SkinThickness, Insulin and BMI were treated as missing data and imputed with the column median before training.

6.2 UCI Heart Disease Dataset

303 records and 14 columns describing age, sex, chest-pain type (cp), resting blood pressure (trestbps), cholesterol (chol), fasting blood sugar (fbs), resting ECG (restecg), maximum heart rate (thalach), exercise-induced angina (exang), ST depression (oldpeak), ST slope (slope), number of major vessels (ca), thalassemia (thal), and the binary target label. Rows with missing values were dropped, leaving 297 complete records for training.

7. Methodology

7.1 Data Preparation

- Zero-as-missing correction on the diabetes dataset, followed by median imputation.
- Removal of incomplete rows from the heart-disease dataset.
- An 80/20 train–test split for both datasets, with a fixed random seed for reproducibility.

7.2 Model Training & Selection

For each screening task, two gradient-boosted / ensemble classifiers were trained and compared head-to-head: Random Forest and XGBoost. Each model was evaluated on Accuracy, ROC-AUC and F1-score on the held-out test

set (see Section 8 for the full results). Random Forest was selected as the deployed model for both diabetes and heart-disease screening, based on its stronger overall balance of the three metrics. The trained models, together with the exact list of feature columns they expect, are serialized with joblib and loaded once per session inside the Streamlit app.

7.3 Symptom & Report Fusion

Two lightweight, rule-based helpers extend the ML pipeline beyond the raw form fields. A keyword-based symptom parser maps phrases such as "frequent urination", "excessive thirst" or "chest pain" to small adjustments on the relevant model features, letting the free-text symptom box gently influence the screening result. A report-reading pipeline extracts text from PDF reports (pypdf) or photographed reports (pytesseract OCR), then a set of regular expressions pulls out common clinical values — glucose, cholesterol, BMI, blood pressure, heart rate — which are merged into the same feature set used by the manual form, with report values taking priority when present.

7.4 Explainability

A SHAP TreeExplainer is used to compute per-prediction feature contributions, so that — for any individual patient record — the top factors pushing the risk score up or down can be surfaced alongside the headline percentage, supporting more transparent human review.

7.5 Conversational Guidance

All natural-language output — the sidebar triage chat and the structured post-screening guidance — is generated by a single Groq-hosted language model ("openai/gpt-oss-20b", used on Groq's free tier). The model is instructed through a fixed system prompt to ask short follow-up questions before giving guidance, to never diagnose or prescribe, to always flag emergency warning signs, and to reply in the language the patient selected (English, Urdu or Roman Urdu).

8. Results & Evaluation

Both candidate models were evaluated on the held-out 20% test split of each dataset. The table below reports Accuracy, ROC-AUC and F1-score for each combination of dataset and algorithm.

Dataset	Model	Accuracy	ROC-AUC	F1-Score
Diabetes	XGBoost	0.7597	0.8270	0.6408
Diabetes	Random Forest (selected)	0.7597	0.8257	0.6726
Heart Disease	XGBoost	0.8333	0.9051	0.8077
Heart Disease	Random Forest (selected)	0.8667	0.9408	0.8519

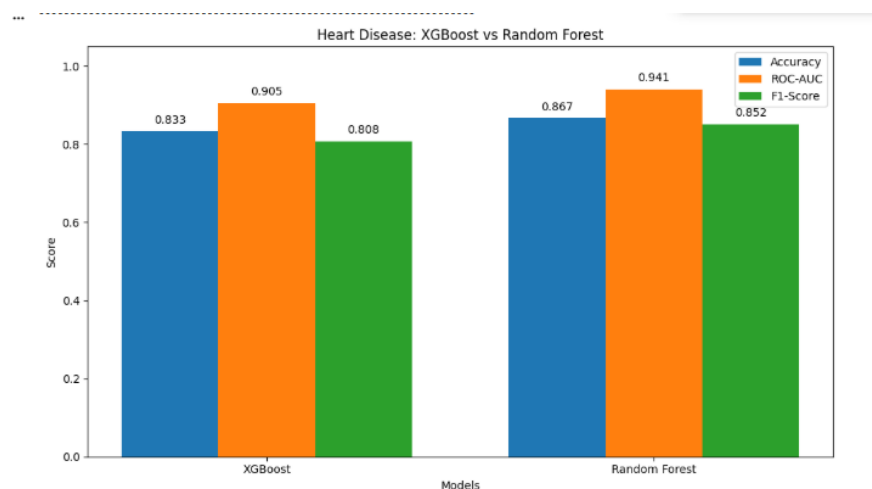


Figure 3 — XGBoost vs. Random Forest comparison on the Heart Disease test set.

For the diabetes model, Random Forest and XGBoost achieved identical accuracy (0.7597), but Random Forest produced a noticeably better F1-score (0.6726 vs. 0.6408), indicating a better balance between precision and recall on the minority (at-risk) class. For the heart-disease model, Random Forest outperformed XGBoost on every metric — accuracy (0.8667 vs. 0.8333), ROC-AUC (0.9408 vs. 0.9051) and F1-score (0.8519 vs. 0.8077). Random Forest was therefore selected as the production model for both screening tasks in the deployed application.

These figures describe the models' performance on public benchmark test data, not on real-world clinical outcomes. In the live application, results are always presented as screening-level estimates and are explicitly not a medical diagnosis.

9. Application Walkthrough

The Patient Screening tab offers two modes so the same underlying models serve both an ordinary patient at home and a health worker with laboratory results in hand.

The screenshot shows the 'Simple Mode' of the SehatAI app. On the left is a sidebar chat with a 'SehatAI Health Assistant' who asks for symptoms and offers quick start options. The main screen is titled 'Rural Health Risk & Triage Copilot' and includes tabs for 'Patient Screening', 'Report Analysis', and 'About SehatAI'. Under 'Patient Information', there's a 'Simple Mode (easy questions for everyone)' toggle. Below this, a text box explains that Simple Mode uses typical values for technical data. The form fields include Patient Name, Age (30), Sex (Male), Weight (65.00 kg), and Height (165.00 cm). A 'Manage app' button is in the bottom right corner.

Figure 4 — Simple Mode: plain-language questions, with a live triage chat in the sidebar.

The screenshot shows the 'Advanced Mode' of the SehatAI app. The sidebar chat is identical to the Simple Mode. The main form is the same but includes additional clinical fields. Under 'Diabetes Screening', there are input fields for Glucose (100.00), Blood Pressure (70.00), Skin Thickness (20.00), Insulin (80.00), BMI (25.00), Diabetes Pedigree (0.67), and Diabetes Age (30). Under 'Cardiovascular Screening', there are input fields for Chest Pain Type (0), Resting Blood Pressure (120.00), Cholesterol (200.00), Fasting Blood Sugar (120), Resting ECG (0), Maximum Heart Rate, Exercise Induced Angina, and ST Depression. A 'Manage app' button is in the bottom right corner.

Figure 5 — Advanced Mode: the full clinical form for clinics with exact lab values.

In Simple Mode, the app asks only what an ordinary person can reasonably know about themselves — weight and height (used to compute BMI), an approximate "120/80"-style blood-pressure reading, and "I don't know" checkboxes that fall back to typical population values for anything unknown. Advanced Mode exposes every clinical field the two models were trained on — glucose, insulin, cholesterol, resting ECG, ST depression, thalassemia, and so on — for health workers who have an actual lab report in front of them. Both modes share the same Report Analysis tab, which reads an uploaded PDF or image report, extracts any recognisable values, runs them through the same Random Forest models, and adds a Groq-generated plain-language explanation of the report.

10. Key Features

- Dual-mode screening form — Simple Mode for everyone, Advanced Mode for clinics with lab data.
- Two trained, evaluated and compared ML risk models (diabetes and cardiovascular disease).

- Automatic PDF/image report reading with regex-based value extraction and model-based re-scoring.
- Conversational sidebar assistant with quick-reply symptom chips and short follow-up questions.
- Full bilingual support across the form, chatbot and AI-generated guidance (English / Urdu / Roman Urdu).
- Consistent, prominent non-diagnostic disclaimers and emergency-warning-sign messaging throughout.

11. Limitations & Safety Considerations

- SehatAI is a decision-support prototype, not a certified medical device, and must not be used as a substitute for professional diagnosis.
- The training datasets (Pima Diabetes, UCI Heart Disease) are decades-old public benchmarks with limited size and demographic diversity, so risk estimates may not generalise perfectly to every population.
- Values not available to the patient are filled in Simple Mode with population-typical defaults, which trades some individual accuracy for accessibility.
- Image-based OCR quality depends on photo clarity and the availability of the tesseract-ocr system package on the hosting platform.
- The conversational assistant is explicitly instructed never to diagnose or prescribe, but, like any language model, its output should still be reviewed by a qualified professional before acting on it.

12. Future Work

- A native Android/iOS companion app with offline-friendly screening for areas with poor connectivity.
- A vision-capable OCR/LLM pipeline for messier, handwritten, or low-quality report photos.
- Additional screening models — anemia, hypertension, malnutrition — trained on more diverse datasets.
- A clinician-facing dashboard with longitudinal patient history, backed by the optional Supabase integration already sketched in the architecture.

13. Conclusion

SehatAI demonstrates that a carefully designed combination of classical machine learning, simple rule-based extraction, and a free-tier large language model can produce a genuinely accessible first-line health screening tool for rural and under-served communities. By offering both a Simple Mode built around what an ordinary patient can know and an Advanced Mode for clinics with lab data, the same trained models serve two very different users without compromising on either accessibility or clinical detail. The project was built and evaluated end-to-end — from public benchmark datasets, through model comparison and selection, to a deployed, bilingual, conversational web application — during the PakAngels GenAI & Agentic AI Training hackathon.

14. Team & Contributions

Name	Role	Primary Contribution
Jahangeer Ali	Team Lead · AI Engineer	Project leadership, ML model design, risk-engine and Streamlit application development.
Kheem Pakrnash Dharmani	AI Engineer	Model training, evaluation and comparison; report-analysis pipeline.
Maria Talib	Civil Engineer	Requirements analysis, documentation and user-scenario design for rural deployment.
Huda Tahir	IT Expert	Testing, deployment support and quality assurance.
Hajra Arshad	Web Developer	Streamlit UI/UX design and bilingual interface implementation.
Saira Lateef	IT Expert	Testing, deployment support and quality assurance.



Figure 6 — Team SehatAI.

15. References

- Pima Indians Diabetes Dataset — jbrownlee/Datasets, GitHub (pima-indians-diabetes.data.csv).
- UCI Machine Learning Repository — Heart Disease Dataset.
- Streamlit documentation — <https://docs.streamlit.io>
- scikit-learn documentation — <https://scikit-learn.org>
- XGBoost documentation — <https://xgboost.readthedocs.io>
- SHAP documentation — <https://shap.readthedocs.io>
- Groq API documentation — <https://console.groq.com/docs>
- PakAngels GenAI & Agentic AI Training, Cohort 11 — Mid-Hackathon Guidelines, in collaboration with iCodeGuru and ASPIRE Pakistan.